A Production Economy

Capital, rational expectations and the Dynare decision rule

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19 November 2026

What changes when households can invest?

Last session

Aggregate consumption was pinned down by the endowment. Household saving decisions determined the equilibrium real rate, and approximation made the decision rule computable.

Today

Saving can build productive capital. How does a temporary productivity shock change consumption and investment, and why do its effects persist?

Our route

Derive the production economy, select its stable solution, then use Dynare to connect policy coefficients to impulse responses and moments.

Plan

1. The production economy
2. Solving the linear model
3. Dynare implementation
4. Extensions and interpretation

From endowments to production: today's saving changes tomorrow's resources.

From equilibrium to dynamics

economic restrictions $\longrightarrow$ stable decision rule $\longrightarrow$ model-implied dynamics.

- Derive the equilibrium, steady state and local approximation.
- Solve a small model by hand to understand what Dynare computes.
- Read policy coefficients, impulse responses and moments as economic objects.
- Carry the resulting transition equation into the likelihood in Session 3.

180 minutes of work, plus a 15-minute break

Production, optimality and steady state	35 min
Linearization and stable decision rule	55 min
Dynare implementation and interpretation	40 min
Extensions, exercise checkpoints and synthesis	50 min

The 180 minutes include 30 minutes of exercise work and discussion. Take a separate 15-minute break after the stable-rule derivation.

Capital and fixed labour

$$Y_{jt} = A_t K_{j,t-1}^{\alpha}, \qquad 0 < \alpha < 1.$$

- Household $j \in [0, 1]$ consumes, produces and accumulates physical capital.
- A_t: common productivity. $K_{j,t-1}$: installed capital available for production at t.
- Labour is inelastic. There is no hours equation to add to this benchmark.
- Diminishing returns to capital limit the incentive to accumulate indefinitely.

Investment carries resources forward

$$C_{jt} + I_{jt} = Y_{jt},$$
$$K_{jt} = (1-\delta)K_{j,t-1} + I_{jt}.$$

- δ: fraction of installed capital that depreciates each period.
- K_{jt} is chosen today, then used for production at $t+1$.
- Combining the two equations gives the single feasibility restriction

$$C_{jt} + K_{jt} = A_t K_{j,t-1}^{\alpha} + (1-\delta)K_{j,t-1}.$$

Consumption today, production tomorrow

$$\max_{\{C_{j,t+\tau},K_{j,t+\tau}\}_{\tau\geq 0}} E_t \sum_{\tau=0}^{\infty} \beta^\tau \log C_{j,t+\tau}.$$

$$\log A_t = \rho_A \log A_{t-1} + \eta_t^A, \qquad \eta_t^A \sim \mathcal{N}(0, \sigma_A^2).$$

- Taking consumption today sacrifices capital and future output.
- $0 < \beta < 1$: patience. $0 \leq \rho_A < 1$: persistence of productivity.
- The innovation η_t^A is not known before date t.

The capital choice spans two dates

$$\mathcal{L} = E_t \sum_{\tau \geq 0} \beta^\tau \left[\log C_{j,t+\tau} + \lambda_{j,t+\tau} \left(A_{t+\tau} K_{j,t+\tau-1}^{\alpha} + (1-\delta) K_{j,t+\tau-1} - C_{j,t+\tau} - K_{j,t+\tau} \right) \right].$$

- Consumption choice: $\lambda_{jt} = 1/C_{jt}$.
- Buying capital costs λ_{jt} today.
- Tomorrow it produces an extra $\alpha A_{t+1} K_{jt}^{\alpha-1}$ units and leaves $1-\delta$ units of capital.
- The capital payoff is uncertain, unlike Session 1's risk-free bond return.

Derive the two first-order conditions

$$\frac{\partial \mathcal{L}}{\partial C_{jt}} = \frac{1}{C_{jt}} - \lambda_{jt} = 0, \qquad \lambda_{jt} = \frac{1}{C_{jt}},$$
$$\frac{\partial \mathcal{L}}{\partial K_{jt}} = -\lambda_{jt} + \beta E_t[\lambda_{j,t+1}(\alpha A_{t+1} K_{jt}^{\alpha-1} + 1 - \delta)] = 0.$$

A unit of capital costs one unit of current resources. Tomorrow it yields its marginal product and its undepreciated part.

The total return to capital

$$\frac{1}{C_{jt}} = \beta E_t \left[\frac{\alpha A_{t+1} K_{jt}^{\alpha-1} + 1 - \delta}{C_{j,t+1}} \right].$$

- Current marginal consumption value equals the expected discounted capital payoff.
- $\alpha A_{t+1} K_{jt}^{\alpha-1}$ is the marginal product of capital.
- $1 - \delta$ is the undepreciated capital carried forward.
- The return and marginal utility are jointly random, so keep their product inside the expectation.

The three equilibrium conditions

Identical households and common productivity imply $C_{jt} = C_t$ and $K_{jt} = K_t$:

$$
\begin{aligned}
C_t + K_t &= A_t K_{t-1}^{\alpha} + (1-\delta) K_{t-1}, \\
\frac{1}{C_t} &= \beta E_t \left[\frac{\alpha A_{t+1} K_t^{\alpha-1} + 1 - \delta}{C_{t+1}} \right], \\
\log A_t &= \rho_A \log A_{t-1} + \eta_t^A, \qquad \eta_t^A \sim \mathcal{N}(0, \sigma_A^2).
\end{aligned}
$$

Feasibility, intertemporal optimality and the shock process must hold jointly. Installed capital is the dated object in production.

Symmetry closes the aggregate model

$$C_t = \int_0^1 C_{jt}\, dj, \qquad K_t = \int_0^1 K_{jt}\, dj.$$

- Identical households face the same shock and have identical initial capital.
- In the symmetric equilibrium, $C_{jt} = C_t$ and $K_{jt} = K_t$.
- Therefore $Y_t = A_t K_{t-1}^{\alpha}$ and the individual Euler condition becomes the aggregate one.
- Without symmetry, $\int K_{jt}^{\alpha} dj$ need not equal K_t^{α}.

Steady-state accounting

At $\bar{A} = 1$ and with zero innovations,

$$\begin{aligned} \bar{C} + \bar{K} &= \bar{K}^{\alpha} + (1-\delta)\bar{K}, \\ \bar{Y} &= \bar{K}^{\alpha}, \qquad \bar{I} = \delta \bar{K}, \qquad \bar{C} = \bar{Y} - \bar{I}, \\ 1 &= \beta[\alpha \bar{K}^{\alpha-1} + 1 - \delta]. \end{aligned}$$

The Euler condition selects capital. Feasibility partitions steady-state output between consumption and replacement investment.

Steady-state capital

$$
\begin{aligned}
1 &= \beta\left[\alpha \bar{K}^{\alpha-1} + 1 - \delta\right], \\
\alpha \bar{K}^{\alpha-1} &= \beta^{-1} - 1 + \delta, \\
\bar{K} &= \left(\frac{\beta^{-1} - 1 + \delta}{\alpha}\right)^{1/(\alpha-1)}.
\end{aligned}
$$

- Set $\bar{A} = 1$, remove date indices and set innovations to zero.
- The total gross return must equal $1/\beta$.
- Greater patience raises the target capital stock because the required return is lower.

Steady-state levels and calibration

$$\bar{Y} = \bar{K}^{\alpha}, \qquad \bar{I} = \delta \bar{K}, \qquad \bar{C} = \bar{Y} - \bar{I}.$$

$\alpha = .40$	$\beta = .99$	$\delta = .025$	$\bar{A} = 1$
$\bar{K} \simeq 57.708$	$\bar{Y} \simeq 5.064$	$\bar{I} \simeq 1.443$	$\bar{C} \simeq 3.621$

- These are model units under the technology normalization, not national-account levels.
- The supplied simulation uses $\rho_A = .85$ and $\sigma_A = .007$.
- The handout's historical console example uses $\rho_A = .95$. Do not mix their numerical outputs.

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The equilibrium system is in place. Select a stable decision rule, not just an algebraic root.

From equilibrium conditions to a linear system

A rational-expectations model is a collection of nonlinear restrictions:

$$E_t[f_\theta(x_{t+1}, x_t, x_{t-1}, \epsilon_t)] = 0.$$

Around a deterministic steady state, first-order approximation gives

$$E_t[\hat{f}_\theta(\hat{x}_{t+1}, \hat{x}_t, \hat{x}_{t-1}, \epsilon_t)] = 0.$$

θ collects structural parameters, x_t are endogenous variables and ϵ_t are Gaussian innovations. Local solution means a rule that satisfies the linearized system for every admissible state and shock.

States and controls

Object at date t	Role	Timing
$\hat{k}_{t-1}$	inherited physical capital	fixed before today's shock
$\hat{a}_t$	exogenous productivity state	observed after today's shock
$\hat{c}_t$	consumption control	adjusts immediately
$\hat{k}_t$	chosen next-period stock	implied by today's allocation

- A jump variable can still be persistent because it responds to persistent states.
- Dynare also treats persistent exogenous processes as endogenous state variables.

Three equation blocks

$$
\begin{aligned}
0 &= A_0 \hat{s}_t + A_{-1} \hat{s}_{t-1} + B_0 \hat{\jmath}_t + C_0 \hat{z}_t, \\
0 &= E_t \big[D_1 \hat{s}_{t+1} + D_0 \hat{s}_t + D_{-1} \hat{s}_{t-1} \\
&\qquad + M_1 \hat{\jmath}_{t+1} + M_0 \hat{\jmath}_t + L_1 \hat{z}_{t+1} + L_0 \hat{z}_t \big], \\
\hat{z}_t &= Z \hat{z}_{t-1} + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, Q).
\end{aligned}
$$

- The first block has no leads. Current variables are known at t.
- The second block includes forward-looking restrictions.
- The last block describes the exogenous processes.

Four matrix restrictions

Use $\hat{s}_t = F_s \hat{s}_{t-1} + G_s \hat{z}_t$ and $\hat{\jmath}_t = F_j \hat{s}_{t-1} + G_j \hat{z}_t$. The two coefficients in the static block must vanish:

$$\begin{aligned} 0 &= A_0 F_s + A_{-1} + B_0 F_j, \\ 0 &= A_0 G_s + B_0 G_j + C_0. \end{aligned}$$

The forward block gives two more restrictions:

$$\begin{aligned} 0 &= D_1 F_s^2 + D_0 F_s + M_1 F_j F_s + D_{-1} + M_0 F_j, \\ 0 &= D_1 F_s G_s + D_0 G_s + M_1 F_j G_s + M_0 G_j \\ &\quad + D_1 G_s Z + M_1 G_j Z + L_1 Z + L_0. \end{aligned}$$

The state restriction is a matrix quadratic. Product order matters.

The stable root pins down the remaining matrices

If B_0 is invertible,

$$F_j = -(B_0)^{-1}(A_0 F_s + A_{-1}).$$

Then the two linear systems for shock responses are

$$A_0 G_s + B_0 G_j = -C_0,$$
$$(D_1 F_s + D_0 + M_1 F_j) G_s + D_1 G_s Z + M_0 G_j + M_1 G_j Z = -L_1 Z - L_0.$$

The displayed order of matrix products is part of the model. Dynare selects the stable invariant subspace rather than asking the user to solve every entry by hand.

Include the exogenous state in the transition

Substitute $\hat{z}_t = Z\hat{z}_{t-1} + \epsilon_t$ into both decision rules:

$$\underbrace{\begin{bmatrix} \hat{s}_t \\ \hat{z}_t \\ \hat{\jmath}_t \end{bmatrix}}_{\hat{x}_t} = \underbrace{\begin{bmatrix} F_s & G_s Z & 0 \\ 0 & Z & 0 \\ F_j & G_j Z & 0 \end{bmatrix}}_{F} \underbrace{\begin{bmatrix} \hat{s}_{t-1} \\ \hat{z}_{t-1} \\ \hat{\jmath}_{t-1} \end{bmatrix}}_{\hat{x}_{t-1}} + \underbrace{\begin{bmatrix} G_s \\ I \\ G_j \end{bmatrix}}_{G} \epsilon_t.$$

- The transition is now driven by innovations, not by the persistent process itself.
- Lagged controls have zero coefficients: inherited states carry their information.
- This is the handout's augmented representation, $\hat{x}_t = F\hat{x}_{t-1} + G\epsilon_t$.

Blanchard–Kahn: roots and rank

- A unique local stable solution requires the appropriate count of unstable generalized eigenvalues and forward-looking directions.
- Too many unstable directions: generally no stable path for arbitrary inherited states.
- Too few: generally multiple stable paths, or indeterminacy.
- The associated rank condition must also hold. A count alone is insufficient.
- Leads in the productivity equation can affect Dynare's reported classification. Read the diagnostics for the actual declared system.
- Diagnostic exercise: change α in a copy. Recompute the steady state and roots rather than assuming that increasing returns imply instability.

Read and work: construct the benchmark

Read Handout 2 §1 and Exercise 1, questions 1–4. Take 10 minutes to reconstruct the objective, capital return and linear resource constraint.

- Check identical initial capital before aggregating nonlinear technology.
- Keep installed capital at $t-1$ in production and chosen capital at t in the Euler condition.
- Use §1.1 as a checkpoint after your derivation, not as a substitute for it.

Checkpoint: linearize the resource constraint

$$C_t + K_t = A_t K_{t-1}^{\alpha} + (1-\delta) K_{t-1}.$$

1. Define $\hat{c}_t = \log(C_t/\bar{C})$, $\hat{k}_t = \log(K_t/\bar{K})$ and $\hat{a}_t = \log A_t$.
2. Subtract the steady state, then apply a first-order expansion term by term.
3. Keep the steady-state coefficients on the consumption and capital terms.
4. Divide by $\bar{K}$ to isolate $\hat{k}_t$.

Check: why is the capital term on the production side dated $t-1$?

Expand the resource constraint

For $\hat{c}_t = \log(C_t/\bar{C})$, $\hat{k}_t = \log(K_t/\bar{K})$ and $\hat{a}_t = \log A_t$,

$$C_t \simeq \bar{C}(1+\hat{c}_t), \qquad K_t \simeq \bar{K}(1+\hat{k}_t),$$
$$A_t K_{t-1}^{\alpha} \simeq \bar{K}^{\alpha}(1+\hat{a}_t+\alpha\hat{k}_{t-1}), \qquad (1-\delta)K_{t-1} \simeq (1-\delta)\bar{K}(1+\hat{k}_{t-1}).$$

Subtract the steady state:

$$\bar{C}\hat{c}_t + \bar{K}\hat{k}_t = \bar{K}^{\alpha}(\hat{a}_t + \alpha\hat{k}_{t-1}) + (1-\delta)\bar{K}\hat{k}_{t-1}.$$

Resource linearization keeps steady-state weights

$$\bar{C}\hat{c}_t + \bar{K}\hat{k}_t = \bar{K}^{\alpha}(\hat{a}_t + \alpha\hat{k}_{t-1}) + (1-\delta)\bar{K}\hat{k}_{t-1},$$
$$\hat{k}_t = k_1\hat{a}_t + k_2\hat{k}_{t-1} - k_3\hat{c}_t.$$

$$k_1 = \bar{K}^{\alpha-1}, \qquad k_2 = \alpha\bar{K}^{\alpha-1} + 1 - \delta, \qquad k_3 = \bar{C}/\bar{K}.$$

- More productivity or inherited capital raises feasible end-of-period capital.
- More consumption lowers investment and end-of-period capital.

The Euler expansion uses the steady-state return

Let $\bar{R} = \alpha \bar{K}^{\alpha-1} + 1 - \delta$. Then

$$R_{t+1} \simeq \bar{R} + \alpha \bar{K}^{\alpha-1}[\hat{a}_{t+1} + (\alpha - 1)\hat{k}_t].$$

Since $1/C_t \simeq (1 - \hat{c}_t)/\bar{C}$ and $\beta \bar{R} = 1$,

$$-\hat{c}_t = E_t \left[-\hat{c}_{t+1} + \beta \alpha \bar{K}^{\alpha-1} (\hat{a}_{t+1} + (\alpha - 1)\hat{k}_t) \right].$$

Only after this expansion do we use $E_t \hat{a}_{t+1} = \rho_A \hat{a}_t$ in the solved linear system.

Euler linearization uses the steady-state return

$$-\hat{c}_t = E_t\left[-\hat{c}_{t+1} + \beta\alpha\bar{K}^{\alpha-1}\left(\hat{a}_{t+1} + (\alpha-1)\hat{k}_t\right)\right],$$

$$\hat{c}_t = E_t\hat{c}_{t+1} - c_1 E_t\hat{a}_{t+1} - c_2\hat{k}_t.$$

$$c_1 = \beta\alpha\bar{K}^{\alpha-1} > 0, \qquad c_2 = c_1(\alpha-1) < 0.$$

- The normalization uses $\beta\bar{R} = 1$, where $\bar{R}$ is the total gross return.
- Higher expected productivity raises the incentive to invest today.
- More capital lowers its future marginal product through diminishing returns.

Three equations organize the entire benchmark

$$\hat{k}_t = k_1 \hat{a}_t + k_2 \hat{k}_{t-1} - k_3 \hat{c}_t,$$
$$\hat{c}_t = E_t \hat{c}_{t+1} - c_1 E_t \hat{a}_{t+1} - c_2 \hat{k}_t,$$
$$\hat{a}_t = \rho_A \hat{a}_{t-1} + \eta_t^A.$$

- Feasibility allocates current resources.
- The Euler equation imposes an intertemporal optimality condition.
- Productivity brings an exogenous source of uncertainty.
- A solution must satisfy all three for every admissible inherited state and innovation.

Guess a linear rule with four unknown coefficients

$$\hat{c}_t = F_c \hat{k}_{t-1} + G_c \hat{a}_t, \qquad \hat{k}_t = F_k \hat{k}_{t-1} + G_k \hat{a}_t.$$

- F_c, F_k: responses to inherited capital.
- G_c, G_k: responses to current productivity.
- The coefficients depend on the structural calibration, not on the realized shock.
- Substitute the rules into the model, then match the coefficients of each independent state.

Two restrictions from feasibility

$$0 = (k_2 - k_3 F_c - F_k)\hat{k}_{t-1} + (k_1 - k_3 G_c - G_k)\hat{a}_t.$$

- The equality must hold for any inherited capital and current productivity.
- The capital coefficient must vanish:

$$F_k = k_2 - k_3 F_c.$$

- The productivity coefficient must also vanish:

$$G_k = k_1 - k_3 G_c.$$

Forecast tomorrow's consumption

$$
\begin{aligned}
E_t \hat{c}_{t+1} &= F_c \hat{k}_t + G_c E_t \hat{a}_{t+1}, \\
E_t \hat{a}_{t+1} &= \rho_A \hat{a}_t, \\
\hat{c}_t &= (F_c - c_2)\hat{k}_t + \rho_A (G_c - c_1)\hat{a}_t.
\end{aligned}
$$

- Today's chosen capital becomes tomorrow's inherited state.
- Future innovations have conditional mean zero.
- Now substitute the rules for $\hat{c}_t$ and $\hat{k}_t$ into the last line.

The last two coefficient restrictions

$$0 = -F_c + (F_c - c_2)F_k,$$
$$0 = -G_c + (F_c - c_2)G_k + \rho_A(G_c - c_1).$$

- Together with the resource restrictions, there are four equations for four coefficients.
- But the term $F_c F_k$ makes the coefficient problem nonlinear.
- Finding an algebraic root is not enough. The implied path must also be stable.

Next step: eliminate F_k and solve a scalar quadratic.

Choose the root that stabilizes capital

$$-k_3 F_c^2 + (k_3 c_2 - 1 + k_2) F_c - k_2 c_2 = 0, \qquad F_k = k_2 - k_3 F_c.$$

Candidate F_c	Implied F_k	Capital dynamics
-0.513485	1.042323	explosive
0.653600	0.969086	stable

- The criterion is $|F_k| < 1$, with $|\rho_A| < 1$ for the productivity state.
- Selecting $0 < F_c < 1$ happens to work here, but is not a general stability test.

Analytical calculation for $\alpha = .40$, $\beta = .99$, $\delta = .025$.

Solve the shock responses

$$G_c = \frac{(F_c - c_2)k_1 - \rho_A c_1}{(F_c - c_2)k_3 + 1 - \rho_A}, \qquad G_k = k_1 - k_3 G_c.$$

- At $\rho_A = .85$: $G_c \simeq 0.154153$ and $G_k \simeq 0.078079$.
- A productivity innovation is split between current consumption and capital accumulation.
- Changing shock persistence changes these coefficients even though the deterministic steady state is unchanged.
- Compare a more persistent shock by re-solving the coefficients, not merely extending the same IRF.

Embed productivity in the transition rule

$$\underbrace{\begin{bmatrix}\hat{k}_t\\ \hat{a}_t\\ \hat{c}_t\end{bmatrix}}_{x_t} = \underbrace{\begin{bmatrix}F_k & G_k\rho_A & 0\\ 0 & \rho_A & 0\\ F_c & G_c\rho_A & 0\end{bmatrix}}_{F} \underbrace{\begin{bmatrix}\hat{k}_{t-1}\\ \hat{a}_{t-1}\\ \hat{c}_{t-1}\end{bmatrix}}_{x_{t-1}} + \underbrace{\begin{bmatrix}G_k\\ 1\\ G_c\end{bmatrix}}_{G} \eta_t^A.$$

- The lagged-consumption column is zero. Consumption has no independent state in the benchmark.
- F propagates past conditions. G maps a new innovation into current variables.
- This is the transition equation needed for simulation and state-space estimation.

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The hand solution becomes a benchmark for checking Dynare output.

From equations to Dynare

1. Declare the three endogenous variables and one innovation.
2. Calibrate structural parameters and compute the steady-state coefficients.
3. Enter the three already-linearized equations with the correct timing.
4. Check residuals and determinacy, then simulate and interpret.

Benchmark file: `session2/codes/basicRBC.mod`. The file `RBC.mod` is a different reduced-form teaching system.

Declarations and calibration

```
var k a c;
varexo e_a;
parameters alpha beta delta rho Css Kss;
alpha = .40;
beta = .99;
delta = .025;
rho = .85;
Kss = ((1/beta-(1-delta))/alpha)^(1/(alpha-1));
Css = Kss^alpha-delta*Kss;
```

- In the file, `rho` is the productivity persistence ρ_A.
- `k`, `a` and `c` will represent log deviations, not levels.

Code the equations and their timing

```
model(linear);
    k = Kss^(alpha-1)*a
        +(alpha*Kss^(alpha-1)+1-delta)*k(-1)
        -Css/Kss*c;
    c = c(+1)-beta*alpha*Kss^(alpha-1)
        *(a(+1)+(alpha-1)*k);
    a = rho*a(-1)+e_a;
end;
```

- `k(-1)` is installed capital. `k` is the stock chosen today.
- Leads encode the forward terms in the expected equilibrium condition.
- `model(linear)` means the equations are already linear, not that Dynare should linearize arbitrary expressions for you.

Set shock units, verify the solution, then simulate

```
shocks;
    var e_a;
    stderr .007;
end;
resid;
steady;
check;
stoch_simul(order=1,irf=30) c k a;
```

- The innovation standard deviation is 0.007 in log units, approximately 0.7%.
- Its variance is $0.007^2 = 0.000049$, not 0.007.
- The zero steady state is correct for the linearized variables. Diagnose before reading an IRF.

Read roots and rank separately

The handout's historical run uses $\rho_A = .95$:

```
EIGENVALUES:  0.9500   0.9691   1.0420   1.262e+17
There are 2 eigenvalue(s) larger than 1 in modulus
for 2 forward-looking variable(s)
The rank condition is verified.
```

A unique stable path needs the appropriate unstable-root count and full rank. This illustration is historical. The supplied simulation files use $\rho_A = .85$.

Read the historical policy table

	c	k	a
k(-1)	0.653600	0.969086	0
a(-1)	0.269310	0.066465	0.950000
e_a	0.283484	0.069963	1.000000

$$\begin{bmatrix} \hat{c}_t \\ \hat{k}_t \\ \hat{a}_t \end{bmatrix} = \begin{bmatrix} 0 & .653600 & .269310 \\ 0 & .969086 & .066465 \\ 0 & 0 & .950000 \end{bmatrix} \begin{bmatrix} \hat{c}_{t-1} \\ \hat{k}_{t-1} \\ \hat{a}_{t-1} \end{bmatrix} + \begin{bmatrix} .283484 \\ .069963 \\ 1 \end{bmatrix} \eta_t^A.$$

The zero first column records that lagged consumption is not an independent state.

Predict the first IRF point

1. Run `dynare basicRBC` from `session2/codes`.
2. Locate the policy table and identify the innovation row.
3. Using $G = (G_k, 1, G_c)'$, calculate the impact of $\eta_1^A = .007$ from the steady state.
4. Convert the resulting log deviations to approximate percent.
5. Predict which variable jumps immediately and which stock builds gradually.

Record: calibration, shock normalization, variable order and IRF horizon.

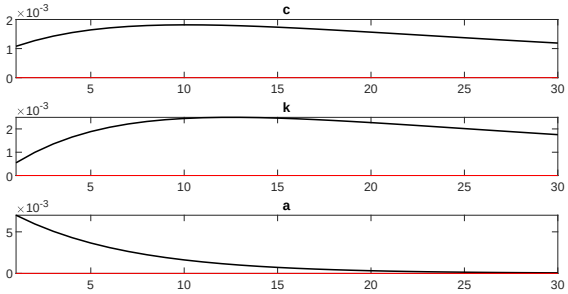
Impact equals policy times shock

$$x_1 = G\eta_1^A = \begin{bmatrix} 0.078079 \\ 1 \\ 0.154153 \end{bmatrix} 0.007 \simeq \begin{bmatrix} 0.000547 \\ 0.007000 \\ 0.001079 \end{bmatrix}.$$

- End-of-period capital increases by approximately 0.0547%.
- Productivity increases by approximately 0.7000%.
- Consumption increases by approximately 0.1079%.
- Installed capital K_0 cannot jump. The first entry concerns the new stock K_1.

Analytical values for the supplied $\rho_A = .85$ calibration. Not an empirical estimate or a new Dynare runtime claim.

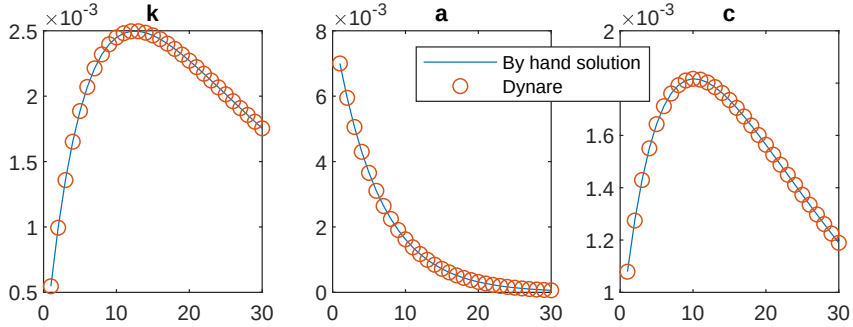
Read the dynamic mechanism, not only the peak



- Productivity jumps, then fades.
- Capital builds through investment.
- Consumption smooths the temporary resource gain.
- Vertical axes are log deviations. A value of .002 is about .2%.

Source: original handout, `imgs/e_a.pdf`. Historical illustration, not regenerated. Do not attach the handout's $\rho_A = .95$ console table to this graphic.

Compare analytical and Dynare IRFs



- A curve from $x_t = Fx_{t-1} + G\eta_t^A$ and Dynare markers should coincide at the same calibration.
- A mismatch points to timing, ordering, shock scale or a coefficient error before it points to economics.

Source: original handout, `compareRBC` figure. Historical illustration, not a fresh execution certificate.

Declaration order differs from rule order

- `M_.endo_names`: variable names in declaration order.
- `oo_.dr.order_var`: declaration indices listed in decision-rule order.
- `oo_.dr.ghx`: responses to the selected state vector, not a full square transition matrix by default.
- `oo_.dr.ghu`: responses to innovations, also in decision-rule order.
- Compare the named `oo_.irfs.k_e_a`, `a_e_a` and `c_e_a` when checking `compareRBC.mod`.

Moments summarize stochastic propagation

$$V = FVF' + G\sigma_A^2 G', \qquad V = \mathrm{Var}(x_t).$$

Analytical quantity, $\rho_A = .85$	capital	productivity	consumption
Standard deviation, log units	.01353	.01329	.00979
Lag-one autocorrelation	.99746	.85000	.99320

- Stocks can be more persistent than the shock because accumulation propagates its effects.
- These are theoretical model moments, not sample statistics or empirical validation.
- For data comparisons, match definitions, frequency, detrending and sample construction.

Correlations and persistence are model implications

Historical handout output, with $\rho_A = .95$:

correlation	$\hat{c}$	$\hat{k}$	$\hat{a}$
$\hat{c}$	1.0000	.9862	.7516
$\hat{k}$	.9862	1.0000	.6319
$\hat{a}$	.7516	.6319	1.0000

lag	$\hat{c}$	$\hat{k}$	$\hat{a}$
1	.9965	.9992	.9500
2	.9917	.9969	.9025
5	.9709	.9823	.7738

Capital is more persistent than consumption in this calibration. These tables are theoretical moments, not evidence from an estimated sample.

Plan

1. The production economy
2. Solving the linear model
3. Dynare implementation
4. Extensions and interpretation

A new mechanism must enter through its equations, not just a renamed parameter.

Read and work: output and investment

Read Handout 2, Exercise 2, questions 1–3. Take 8 minutes to derive the accounting additions before running a working copy.

$$Y_t = A_t K_{t-1}^{\alpha}, \qquad I_t = K_t - (1-\delta)K_{t-1}.$$

1. Derive $\bar{Y}$ and $\bar{I}$.
2. Log-linearize both equations without changing the capital timing.
3. Add the two variables and equations to a working copy of the benchmark.
4. Predict how investment's impact response compares with consumption's response.

Investment is a difference of capital stocks

$$\hat{y}_t = \hat{a}_t + \alpha \hat{k}_{t-1}, \qquad \hat{i}_t = \frac{\hat{k}_t - (1-\delta)\hat{k}_{t-1}}{\delta}.$$

- Use $\bar{I} = \delta \bar{K}$ when dividing the linearized accumulation equation.
- On impact, inherited capital has not changed, so $\hat{y}_1 = .007$.
- With the benchmark coefficients, $\hat{i}_1 \simeq .000547/.025 = .02186$, approximately 2.19%.
- Investment absorbs much of the temporary resource gain. Its response is large relative to its own steady-state level.

Resource shares expose benchmark limits

$$\frac{\bar{I}}{\bar{Y}} = \delta \bar{K}^{1-\alpha} \simeq .285, \qquad \frac{\bar{C}}{\bar{Y}} \simeq .715.$$

- In this closed benchmark, consumption and investment shares must sum to one.
- The handout's rough OECD comparison uses shares near .60 and .20. It is a motivating benchmark, not a new data estimate.
- Those two data shares do not exhaust GDP. Government spending and external balances matter for the comparison.
- A model extension should repair an identified mechanism or accounting gap, not merely add variables.

Read and work: two mechanisms to extend

Handout 2, Exercises 3 and 4

Read both exercises. Use 12 minutes to contrast external and internal habits, then predict a government-spending innovation on impact.

- Exercise 3, questions 1(a–d) and 2(a–d): identify whose lagged consumption enters utility and which derivative changes.
- Exercise 4, questions 1–5: identify the steady-state resource loss, impact accounting and variance-decomposition assumptions.
- Complete the implementations after class. These extensions are not already implemented in the benchmark files.

External habits use a lagged reference

With the benchmark log utility,

$$u(C_{jt} - hC_{t-1}), \qquad 0 \le h < 1, \qquad \lambda_{jt} = \frac{1}{C_{jt} - hC_{t-1}}.$$

- The household takes aggregate past consumption C_{t-1} as given.
- The reference has no individual j subscript: it is external to the household's choice.
- A higher reference level raises current marginal utility at a given C_{jt}.
- In a symmetric steady state, $\bar{\lambda} = 1/[(1-h)\bar{C}]$.

Handout extension. Abel (1990), Constantinides (1990).

Linearize external habits

$$x_t = C_t - hC_{t-1}, \qquad \hat{x}_t = \frac{\hat{c}_t - h\hat{c}_{t-1}}{1-h}, \qquad \hat{\lambda}_t = -\hat{x}_t.$$

- x_t: consumption above the habit reference. It must remain positive.
- The capital Euler equation now uses $\lambda_t = \beta E_t[\lambda_{t+1} R_{t+1}]$.
- Lagged consumption becomes a state, so the previous three-variable decision rule must be re-solved.
- Exercise: compare IRFs for $h = 0$ and $h > 0$. Explain the change in consumption adjustment, rather than asserting a universal ranking.

Internal habits: a future marginal cost

$$x_{jt} = C_{jt} - hC_{j,t-1}, \qquad \lambda_{jt} = \frac{1}{x_{jt}} - \beta h E_t \left[\frac{1}{x_{j,t+1}} \right].$$

- Today's consumption raises tomorrow's own reference level.
- The household internalizes that effect, unlike under external habits.
- The second term subtracts the discounted future marginal cost.
- In steady state, $\bar{\lambda} = (1 - \beta h)/[(1 - h)\bar{C}]$.

Internal habits add lags and leads

$$\hat{\lambda}_t = \frac{-\hat{x}_t + \beta h E_t \hat{x}_{t+1}}{1 - \beta h}, \qquad \hat{x}_t = \frac{\hat{c}_t - h\hat{c}_{t-1}}{1 - h}.$$

- Linearize the two marginal-utility terms before dividing by $\bar{\lambda}$.
- For $h = 0$, recover $\hat{\lambda}_t = -\hat{c}_t$, the original log-utility model.
- For $h > 0$, expected future surplus consumption matters directly.
- Keep feasibility and the capital return unchanged, but update the optimality block and rerun determinacy checks.

Government spending withdraws private resources

$$P_t G_t = T_t, \qquad C_t + I_t = Y_t - \frac{T_t}{P_t}, \qquad Y_t = C_t + I_t + G_t.$$

- G_t: real public spending. T_t: aggregate nominal lump-sum tax. P_t: price level.
- The budget is balanced, so real taxes equal current government purchases.
- Spending does not directly enter the household's utility or production in this extension.
- This is a resource withdrawal from the private sector, not an assumed Keynesian multiplier.

Spending changes resources, not the Euler condition

$$G_t = g^y \bar{Y} \varepsilon_t^G, \qquad \log \varepsilon_t^G = \rho_G \log \varepsilon_{t-1}^G + \eta_t^G.$$
$$\bar{G} = g^y \bar{Y}, \qquad \bar{C} = (1 - g^y)\bar{Y} - \delta \bar{K}.$$

- Handout calibration: $g^y = .20$, $\rho_G = .97$, $\sigma_G = .053$.
- $\bar{\varepsilon}^G = 1$. The shock is a proportional deviation of spending around its own steady state.
- The steady-state Euler equation still determines $\bar{K}$. Consumption is lower.
- At this calibration, $\bar{C}/\bar{Y} \simeq .515$ and $\bar{I}/\bar{Y} \simeq .285$. Government alone does not match every target.

Predict a spending shock before the IRF

$$\bar{Y}\hat{y}_t = \bar{C}\hat{c}_t + \bar{I}\hat{\imath}_t + \bar{G}\hat{g}_t.$$

- For a pure spending innovation, A_t and inherited K_{t-1} are unchanged. Output cannot jump on impact in this fixed-labour model.
- Private consumption and investment must jointly absorb the additional government demand.
- A forward-looking household recognizes the loss of disposable resources from lump-sum taxation.
- Exercise: solve the extension, plot all four resource components and check their weighted adding-up condition.

Variance decomposition depends on calibration

- With independent technology and spending innovations, total variance is the sum of their propagated contributions.
- Changing a shock's variance or persistence changes its contribution, even if preferences are unchanged.
- Ask whether output is more volatile than consumption under the same transformation as the data.
- Lump-sum-tax wealth effects are not a test of Ricardian equivalence. The latter concerns tax versus debt timing, absent from this balanced-budget experiment.
- An impact accounting restriction is not evidence of an empirically estimated fiscal multiplier.

Use the right file for the right question

- `basicRBC.mod`: the three-equation production benchmark, $\beta = .99$, $\rho_A = .85$.
- `compareRBC.mod`: the same calibration plus a hand-derived rule and IRF comparison.
- It selects the unique root satisfying the capital-transition condition $|F_k| < 1$.
- `RBC.mod`: a separate reduced-form linear system. Its parameter named `beta`, set to 1.05, is *not* the household discount factor.
- Habits and government spending are extension exercises, not already implemented claims about these three supplied files.

Read the handout and complete the exercises

- Read Handout 2 §2.1–2.2 for the full matrix substitution and scalar stable-root derivation.
- Use §3–3.1 to reproduce the Dynare workflow. Record calibration, roots, rank and variable ordering.
- Complete Exercise 1 (questions 1–4), Exercise 2 (1–3), Exercise 3 (both habit variants) and Exercise 4 (1–5).
- Bring one check of the analytical versus Dynare IRFs and one explanation of how an extension changes states or optimality.

Takeaways: capital propagates shocks

1. The production benchmark combines feasibility, the capital Euler equation and productivity dynamics.
2. Log-linearization preserves steady-state weights and the timing of installed capital.
3. A stable rule must satisfy the equilibrium restrictions, not just an algebraic root equation.
4. Dynare automates a solution that should remain economically interpretable.
5. Extensions change states, optimality conditions or resource use. They require a new solution and new diagnostics.

Next: from a solved model to data

$$x_t = Fx_{t-1} + G\epsilon_t \qquad \longrightarrow \qquad \text{observations, prediction errors and likelihood.}$$

- We can propagate latent states and structural shocks.
- We still need to decide which model variables correspond to measured series.
- Measurement equations and the Kalman filter connect this transition rule to data.
- Before moving on, retain the equations, calibration, units and verified solution diagnostics.